SINGAPORE UNIVERSITY OF SOCIAL SCIENCES

ICT330

Examination - July Semester 2023

Database Management Systems

Saturday, 18 November 2023 01:00 pm - 03:00 pm

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This examination contains **4** questions and comprises **7** printed pages (including cover page).
2. You must answer **4** questions.
3. All answers must be written in the answer book.
4. This is **Open-book** examination.

At the end of the examination.
Please ensure that you have written your examination number on each answer book used.

Failure to do so will mean that your work cannot be identified.
If you have used more than one answer book, please tie them together with the string provided.

THE UNIVERSITY RESERVES THE RIGHT NOT TO MARK YOUR SCRIPT IF YOU FAIL TO FOLLOW THESE INSTRUCTIONS.

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(Full marks: 100)

Question 1

A bank issues credit cards to its customers. A customer may be issued multiple credit cards which are either primary or supplementary credit cards. Supplementary cards are usually issued to family members or friends of a customer.

A customer initiates an application for a primary credit card with details such as customer name, date of birth, address, annual income, contact and email. If supplementary cards are required, the application includes the application(s) for one or more supplementary cards to be linked to the primary credit card being applied. Each supplementary card application requires the name, date of birth and address of the applicant for a supplementary card.

When approved, the customer is given a customer id. Each credit card, whether primary or supplementary credit cards will have a unique card number, an expiry date, a credit limit. The primary and supplementary cards are then delivered by post to the customer who is responsible for all transactions made under the primary credit card as well as those made under the supplementary cards, if any. As such, outstanding or unpaid amount is reflected with the primary card.

A transaction made using a credit card will record the credit card number, merchant name (e.g., NTUC Fairprice, Tampines Mall), date and time of transaction and the transacted amount. When the transaction is fully processed, a post date is added to the transaction. New transactions with posted date added will be included in the next bill that will be sent to the customer.

Each bill contains a list of transactions incurred by both the primary card and supplementary cards, if any. A bill also includes a statement date, previous outstanding amount, a minimum amount payable and the billing period (start date and end date).

Question 1a

Construct **ONE (1)** conceptual (ER) model from the statements of requirements to represent the data model showing:

Question 1a(i)

entities, with entities name, relevant attributes and identifier,

(12 marks)

Question 1a(ii)

relationships with maximum and minimum cardinalities and relationship name.

(8 marks)

Question 2

A paint company has installed paint mixing machines in its outlets. When a customer chooses a colour code, a service staff will reference a table for the amounts of colour pigments to produce the chosen colour code. He then administers the colour pigments into a standard sized tin of base (white) colour. The paint mixing machine then mixes the base colour with the colour pigments.

Due to manufacturing differences, more (or less) amount of colour pigments may be actually used to get a better match to the chosen colour. For example, the reference table may state that colour code XYZ require n1 amount of colour pigment A and n2 amount of colour pigment B, but the actual amount of colour pigment A used could be m1 and the actual amount of colour pigment B used could be m2.

As colour pigments are expensive, it is important to track the amount of colour pigments used for the various colour codes in a purchase.

You are given the table:
PaintMixing (colourCode, colourPigment, amtPigmentRequired, purchaseId, outletCode, outletAddress, amtPigmentUsed)

Note that

- amtPigmentRequired is the amount of colour pigment required as specified in the reference table for a particular colour of a standard sized tin,
- amtPigmentUsed is the amount of colour pigment used in a particular purchase for a particular colour of a standard sized tin.
- An outlet is identified by outletCode.
- A purchase is identified by purchaseId.
- There is no need to track the number of cans purchased in this question.

State your assumption(s) for only data requirements that are not specified.

Question 2a

Formulate and list the functional and multi-valued dependencies.

(6 marks)

Question 2b

Draw dependency graph and propose candidate key(s) for the table.

(3 marks)

Question 2c(i)

For each MVD in answer to Q2(a), if it is subsumed, identify the FD that subsumes it.

(2 marks)

Question 2c(ii)

Normalise the table to BCNF and 4NF by applying **the BCNF approach** covered in the course text.

Show how you apply the steps to arrive at the tables, explain and discuss whether the structure of each derived table at each iteration is already in BCNF and 4NF (if applicable).

(7 marks)

Question 2d

Assemble your BCNF and 4NF tables and constraints clearly by giving each table and the referential integrity constraint(s), if any, in the forms described below:

Tables in this form:

TABLE-NAME(col-name$_1$, col-name$_2$, ..., col-name$_n$)

- indicate (PK) against the primary key and
- indicate (FK) against the foreign keys, if any.

Referential integrity constraint(s), if any, **together with the child table** in this form:

- TABLE-NAME$_x$.col-name$_{x1}$ ...col-name$_{xn}$ must exist in
- TABLE-NAME$_y$.col-name$_{y1}$...col-name$_{yn}$

(7 marks)

Question 3

Question 3a(i)

Transform the ERD in Figure Q3(a)(i) into a logical model. Indicate foreign keys together with the foreign key constraints that specify the table being referenced in the constraint.

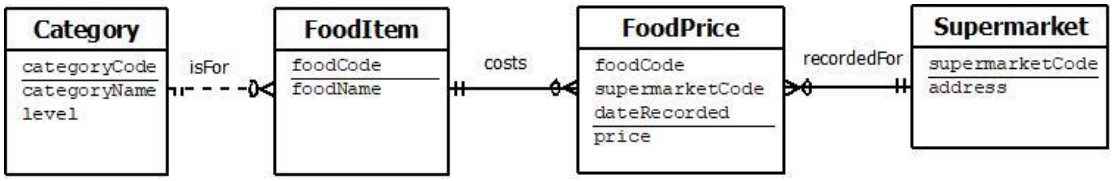

Figure Q3(a)(i)

Tables in this form:
TABLE-NAME(col-name$_1$, col-name$_2$, ..., col-name$_n$)

- indicate (PK) against the primary key and
- indicate (FK) against the foreign keys, if any.

Referential integrity constraint(s), if any, **together with the child table** in this form:
TABLE-NAME$_x$.col-name$_{x1}$ ...col-name$_{xn}$ must exist in
TABLE-NAME$_y$.col-name$_{y1}$...col-name$_{yn}$

(4 marks)

Question 3a(ii)

Refer to the description of the columns for the column definition, and then write the appropriate SQL statements to implement ONLY the tables derived for the **FoodItem** and **FoodPrice** entities in the ERD in Figure Q3(a)(i).

List your SQL statements in the order that they should be executed.

Note that all columns are required. Description for columns in the tables:

- categoryCode is a whole number starting from 1 with step increment 1 and is auto-generated.
- categoryName is at most 20 characters long.
- level is an integer.
- foodCode is exactly 3 character long, and contains an alphabet followed by 2 digits.
- foodName is at most 25 characters long and is unique.
- supermarketCode is 5 characters long.
- dateRecorded is a datetime with default value today's date. dateRecorded cannot be later than today's date.
- price is a decimal number with 5 digits of which 2 are decimal digits. It has a default value 2, and its value is between 1 and 120.
- address is at most 50 characters long.

Refer to Figure Q3(a)(i) for the minimum and maximum cardinalities.

- isFor relationship – both delete and update are disallowed.
- costs relationship – both delete and update are cascaded.
- recordedFor relationship – delete is disallowed and update is cascaded.

(6 marks)

Question 3b

Assume data rows are already in the database.
Formulate SQL queries to display the required resultsets (partial resultset shown as example output):

Question 3b(i)

Display the details of food items with food name that contains the word oil, in ascending order of food code.

foodCode	foodName
O01	Vegetable Oil
O02	Sunflower Oil

(3 marks)

Question 3b(ii)

Display the food code of food items with level 1. Use a subquery. You must not use any Join operation.

foodcode
F07
F08
M01
M02
V01
V02

(6 marks)

Question 3b(iii)

Display supermarket code for supermarkets with more than 5 prices recorded for the month of March in 2022. Show supermarket code together with the number of food prices recorded for the month of March in 2022.

Display the rows in descending order of the number of food prices recorded as well as in ascending order of supermarket code if the number or prices recorded is the same.

supermarketCode	priceCount
MEGAC	11
SUPAA	11

(6 marks)

Question 3c

Implement a trigger to prevent recording more than 15 food prices for any supermarket on any single month of a year. Print a message for the outcome to indicate either success or failure when there is an attempt to record food prices.

(10 marks)

Question 4

Question 4a

Demonstrate the basic techniques of concurrency control and their application.

A watch company has several branches. As a service to customers who are unable buy a watch model that is out of stock in a particular branch, a branch staff may log into the system to locate another branch that has stock of the required model and to make an inter-branch delivery request on behalf of the customer. The stock level of the watch model to be transferred over at the other branch should be reduced accordingly. Explain whether there can be deadlock and/or lost update arising when branch staff from different branches may concurrently access the system to provide this service.

(8 marks)

Question 4b

Describe **ONE (1)** permission that should be given to branch staff to allow them to make an inter-branch delivery request. Describe one permission that should not be given to branch staff.

(2 marks)

Question 4c

Given Figure Q4(c), explain what the database recovery procedure will do for the transactions T1 to T3. Explain also how the transaction log help ensure the ACID properties of T1 to T3.

Record Pointer	Transaction ID	Reverse Pointer	Forward Pointer	Time	Type of Operation	Table	Row ID	Before Image	After Image
1	T1	0	2	10:30:32	Start				
2	T1	1	3	10:30:36	Insert	Table1	1		1, 100, 3
3	T1	2	10	10:30:55	Update	Table2	3	3, 200	3, 250
4	T2	0	5	10:31:01	Start				
5	T2	4	6	10:31:24	Insert	Table1	2		2, 400, 1
6	T2	5	7	10:32:41	Update	Table2	4	4, 300	4, 280
7	T2	6	0	10:32:47	Rollback				
Checkpoint									
8	T3	0	9	10:45:23	Start				
9	T3	8	11	10:46:00	Update	Table2	2	2, 100	2, 200
10	T1	3	0	10:53:33	Rollback				
11	T3	9	12	10:54:02	Update	Table3	2	2, 300, 300	2, 400, 250
12	T3	11	0	10:56:25	Commit				
Crash!									

Figure Q4(c)

(10 marks)

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